Dialect Preservation or Normalization? A Cross-Paradigm Study of German Dialect Transcription

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Abstract

We examine how modern speech foundation models transcribe dialectal speech along a dialect–standard continuum. Using SwissDial (Swiss German) and Betthupferl (Southern German dialects), we shift the focus from conventional accuracy reporting to a behavioral characterization of dialect preservation versus normalization. We introduce Dialect Preference Score (DPS), defined as the difference between normalized character-level similarity to dialect and standard references. Across CTC-based, generative encoder–decoder, and LLM-ASR systems, we observe three regimes: (i) CTC models tend to preserve dialectal forms, (ii) generative models systematically normalize toward standard German, and (iii) LLM-ASR models exhibit a near-zero DPS that masks a high-similarity pseudo-equilibrium between the two references. These results suggest that dialectal behavior is shaped by decoding priors, tokenization, and training distributions rather than by recognition accuracy alone.

1 Introduction

Speech recognition for dialectal varieties remains difficult due to the lack of standardized orthography and substantial variation in phonetic realization. Prior work on dialectal ASR has primarily emphasized improving recognition performance, often treating deviations from standard forms as errors. However, for dialect documentation and linguistic analysis, the central question is not only whether a model is accurate, but also *what kind of text it prefers to produce* when dialect and standard forms compete.

This paper studies dialect preservation and normalization as a behavioral property of modern speech foundation models. Specifically, we ask whether these systems transcribe dialects as distinct linguistic forms or systematically rewrite them into a standard variety. We conduct experiments on two datasets with complementary structure: SwissDial, which provides parallel dialect and Standard High German references (Dogan-Schonberger et al., 2021), and Betthupferl, a historical Southern German benchmark used for robustness validation (Blaschke et al., 2025). Across model paradigms, we find a consistent architecture-linked phase separation: CTC models tend to remain acoustically faithful, whereas autoregressive decoders increasingly default to standard forms, consistent with the "default to standard" behavior documented for text generation systems (Fleisig et al., 2024). LLM-ASR systems occupy an intermediate regime that produces mixed-form outputs rather than purely dialectal or purely standard transcriptions.

2 Methodology

2.1 Datasets

SwissDial. SwissDial is a parallel corpus of Swiss German dialect speech paired with Standard High German references (Dogan-Schonberger et al., 2021). Each utterance includes (i) a dialectal transcription and (ii) a standard reference aligned to the same audio, enabling direct comparison between dialect-preserving and normalizing outputs.

Betthupferl. Betthupferl is an evaluation dataset containing approximately four hours of read speech from multiple dialect groups in Southeast Germany (Blaschke et al., 2025). Unlike SwissDial, Betthupferl does not provide a strictly parallel dialect–standard pair for each utterance; therefore, we use it as a relative validator to test whether paradigm-level ordering replicates under heterogeneous annotation conventions.

2.2 Models

We evaluate three architectural families: (1) **CTC-based models:** wav2vec2 XLS-R (300M, 1B). (2) **Generative ASR models:** Whisper (Tiny to Large-v3). (3) **LLM-ASR models:** Qwen3-ASR (0.6B,

1.7B). These families represent distinct decoding priors, ranging from primarily acoustic inference (CTC) to strong autoregressive language modeling (Whisper, Qwen).

2.3 Metric: Dialect Preference Score (DPS)

Let P denote the predicted transcription, T_{dial} the dialect reference, and T_{std} the standard reference. We define normalized character-level similarities via Levenshtein distance:

$$\text{Sim}_{\text{dial}} = 1 - \frac{\text{Edit}(P, T_{\text{dial}})}{\max(|P|, |T_{\text{dial}}|)}, \quad (1)$$

$$\text{Sim}_{\text{std}} = 1 - \frac{\text{Edit}(P, T_{\text{std}})}{\max(|P|, |T_{\text{std}}|)}, \quad (2)$$

where $\text{Edit}(\cdot, \cdot)$ denotes character-level Levenshtein distance. The Dialect Preference Score is:

$$\text{DPS} = \text{Sim}_{\text{dial}} - \text{Sim}_{\text{std}}. \quad (3)$$

A positive DPS indicates that the output is closer to the dialect reference than to the standard reference; a negative DPS indicates standardization bias.

Collapse Filter. To reduce survival bias under extreme noise, we exclude utterances satisfying:

$$\max(\text{Sim}_{\text{dial}}, \text{Sim}_{\text{std}}) < 0.1. \quad (4)$$

This threshold removes degenerate outputs (for example, repetitive sequences or empty strings) that would otherwise yield misleadingly neutral DPS values.

2.4 Noise Setting

For SwissDial, we evaluate additive Gaussian noise at SNR levels $\{20, 10, 0\}$ dB to probe how decoding priors interact with degraded acoustic evidence.

3 Results

3.1 Paradigm Gap: CTC vs. Generative Models

Figure 1 shows a step-wise shift in DPS distributions across paradigms on SwissDial. CTC models tend to produce positive DPS, consistent with acoustic faithfulness in the absence of a strong language-model prior. In contrast, generative models exhibit negative DPS, indicating systematic rewriting toward standard forms. Within the Whisper family, scaling intensifies the negative drift, suggesting that increased autoregressive capacity amplifies reliance on standard-dominant training distributions.

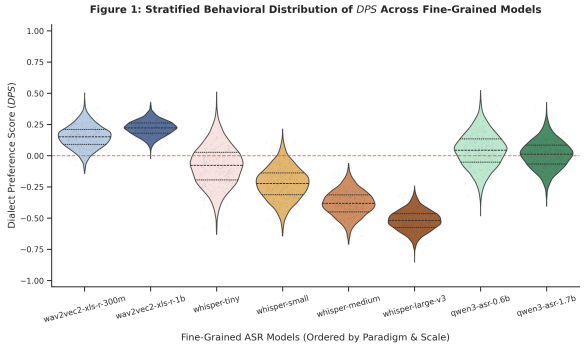


Figure 1: Stratified behavioral distribution of Dialect Preference Score (DPS) across fine-grained models on SwissDial. The violin plots reveal a step-wise downward shift from CTC to Whisper to LLM-ASR paradigms, with scaling within Whisper systematically intensifying standardization bias.

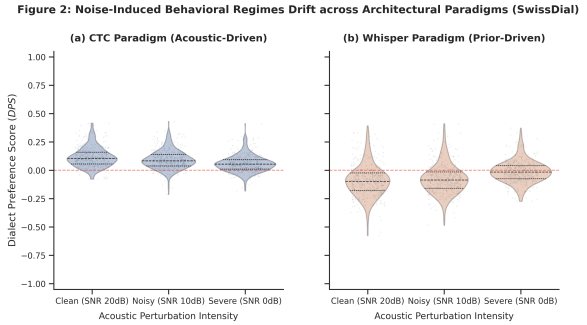


Figure 2: Noise-induced regime drift under progressive acoustic degradation (SNR 20dB, 10dB, 0dB). CTC models maintain positive DPS with increased variance, while generative models exhibit monotonic negative drift, indicating amplified reliance on language model priors when acoustic evidence degrades.

3.2 Noise Robustness and Regime Drift

Under progressive acoustic degradation, CTC models maintain positive DPS with increased variance, whereas generative models exhibit monotonic negative drift (Figure 2). This pattern is consistent with an uncertainty-driven shift toward language-model priors when the acoustic signal becomes less informative.

3.3 Hybrid Behavior in LLM-ASR

LLM-ASR models exhibit near-zero DPS, which can be misread as neutrality. Instead, this behavior reflects a pseudo-equilibrium in which the output remains simultaneously similar to both references, often by producing mixed-form strings that partially preserve dialectal tokens while introducing standard equivalents. This regime resembles a compromise solution rather than an absence of

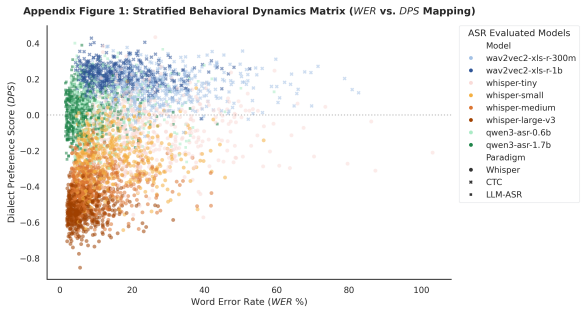


Figure 3: Correlation between Word Error Rate (WER) and Dialect Preference Score (DPS). Each point is color-coded by model. The scatter matrix shows that WER alone does not predict DPS; instead, behavioral trajectories are bounded by architectural paradigm.

preference, and it emphasizes that DPS should be interpreted jointly with absolute similarity levels.

4 Discussion

The results indicate that dialect preservation is not a binary model property, but a continuum shaped by architectural priors. Autoregressive decoders can overwrite dialectal surface forms even when acoustic evidence is present, aligning with the broader observation that high-resource standards often dominate generation behavior (Fleisig et al., 2024). These findings motivate evaluation protocols that separate *recognition quality* from *normalization tendency*, because standard error metrics can conflate dialectal variation with transcription failure.

5 Conclusion

We present a cross-paradigm analysis of dialect transcription behavior in modern speech foundation models. Using DPS, we identify consistent regime differences across CTC, generative, and LLM-ASR systems, and we show that acoustic noise amplifies the reliance on language-model priors. The proposed behavioral framing complements conventional accuracy reporting and supports more faithful dialect documentation.

A Additional Results

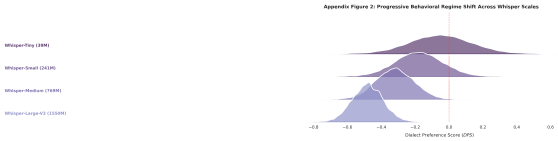


Figure 4: Ridge plot of DPS distribution across Whisper scaling spectrum (Tiny to Large-v3). The continuous leftward shift demonstrates that increasing model capacity within the autoregressive paradigm progressively overwrites dialectal tokens toward standard forms.

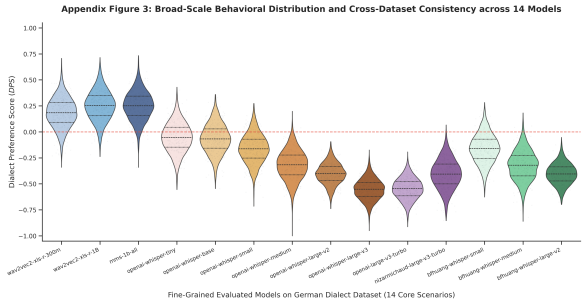


Figure 5: Cross-dataset validation on Betthupferl across 14 models. The architectural phase separation replicates: CTC models show positive DPS retention, while Whisper and its variants maintain negative standardization drift.

References

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